A Decision Support System for Prioritising Industrial Maintenance Interventions

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Abstract. Industrial plants often have more pending maintenance interventions than technicians, spare parts and downtime windows can absorb, so their order has to be decided. Following the Kitchenham guidelines and the PRISMA 2020 statement, 554 records from IEEE Xplore and Web of Science (2021–2026) were screened and 31 studies with evaluation in a real setting were analysed for their criteria, methods, system architecture and validation. Resource constraints, failure likelihood and production impact dominate the criteria, while safety is explicit in only seven studies and is usually traded off against cost. Few proposals are implemented systems, priorities are rarely updated or explained, and only six were tested in the plant. We then propose a decision support system that combines machine learning failure prediction with AHP and TOPSIS, treats safety as a non-compensable constraint and explains each position in the ranking, and characterise the public dataset chosen for its prototype.

Keywords: Industrial maintenance · Decision support systems · Maintenance prioritisation · Multi-criteria decision-making · Predictive maintenance.

1 Introduction

Maintenance managers rarely face a single failure at a time. On a typical day, corrective work orders, preventive tasks and alerts from condition monitoring compete for the same technicians, spare parts and production stoppages, and the order in which they are handled affects safety, output and cost. In many plants this order is still set by experience or by simple urgency codes in the maintenance management system, which makes it hard to justify and to apply consistently across shifts and departments [1, 14].

The spread of sensors and machine learning in Industry 4.0 has improved the ability to predict failures, but a prediction does not tell the manager which intervention to perform first when several are pending. Turning failure predictions, costs, resources and expert knowledge into a justified order of interventions is a multi-criteria decision problem, and decision support systems (DSS) are a natural way to address it.

This paper has two objectives. The first is to review how recent studies prioritise industrial maintenance interventions: which criteria they use, which methods turn those criteria into a priority order, how the resulting systems are built and how they were validated. The second is to use the gaps found in that review to propose the conceptual

architecture of a DSS for this problem and to characterise the dataset on which its prototype will be built.

The paper is organised as follows. Section 2 describes the review method and Section 3 introduces the main maintenance strategies. Section 4 analyses the prioritisation criteria and methods, Section 5 the architecture and validation of the systems, and Section 6 compares the approaches and identifies gaps and trends. Section 7 presents the proposed system and the selected dataset, and Section 8 concludes. Appendix A gathers supporting tables and figures, and Appendix B reports the exploratory analysis of the dataset.

2 Review Methodology

This review follows a structured state-of-the-art approach based on PRISMA 2020 [20]. The robustness of each study was assessed through the validation and limitations recorded during data extraction. Four research questions (RQ) guided the review:

- **RQ1:** Which criteria are considered when prioritising industrial maintenance interventions?
- **RQ2:** Which methods (multi-criteria decision-making, optimisation, machine learning) are used to prioritise these interventions?
- **RQ3:** How are maintenance decision support systems designed in terms of architecture, data sources and interaction with the decision-maker?
- **RQ4:** How are the proposed approaches validated, and what limitations do they present?

2.1 Search Strategy and Eligibility Criteria

IEEE Xplore and the Web of Science Core Collection were searched; Scopus was not available through the institutional subscription. The search string combines three groups of concepts: maintenance, prioritisation and decision-making, and the industrial context. A first version, which applied all terms to every field, returned about 1500 records in IEEE Xplore, many of them only because journal titles matched the industrial terms. The final version therefore requires *maintenance* in the title and the other concepts in the abstract:

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IEEE Xplore: ("Document Title":maintenance) AND ("Abstract":prioriti* OR
"Abstract":priority OR "Abstract":criticality OR "Abstract":"decision support" OR
"Abstract":"multi-criteria" OR "Abstract":AHP OR "Abstract":TOPSIS) AND
("Abstract":industr* OR "Abstract":manufactur* OR "Abstract":equipment)

Web of Science: TI=(maintenance) AND AB=(prioriti* OR priority OR criticality OR
"decision support" OR "multi-criteria" OR AHP OR TOPSIS) AND AB=(industr* OR
manufactur* OR equipment) AND PY=(2021-2026)
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The search string was developed and tested from 15 October 2026, and the final searches were run on 19 October 2026, limited to 2021–2026, one study published online in 2026 appears in a 2027 journal issue [17]. In Web of Science, results were limited to articles, proceedings papers and early access items in English, within ten

engineering, operations research, management and computer science categories, which reduced them from 582 to 385. IEEE Xplore returned 169 records.

A study was included if it was a peer-reviewed journal or conference paper, published between 2021 and 2026, written in English, proposing or applying a prioritisation or decision method for maintenance interventions in an industrial context, and presenting some form of evaluation. Studies were excluded for any of the reasons listed in Table 1 (Appendix A). The exclusion criteria were refined iteratively as the review progressed, with E8, E9 and E10 added to focus it on its central problem and on evidence from real settings. All criteria were applied in the same way to every study.

2.2 Study Selection and Data Extraction

The selection is summarised in the PRISMA diagram of Fig. 1 (Appendix A). Of the 554 records retrieved, 45 duplicates were removed by DOI and normalised title. The remaining 509 were screened by title and abstract, a stage that excluded 233 records, mostly under E8 ($n = 108$) and E3 ($n = 59$). Of the 276 reports sought, 47 could not be accessed (E7). After full-text assessment of the remaining 229, a further 198 were excluded, mainly under E9 ($n = 60$) and E10 ($n = 62$), leaving 31 studies.

For each included study, an extraction form recorded the sector, the prioritisation criteria (RQ1), the decision method and the origin of the weights (RQ2), the architecture and data sources (RQ3), the validation design and results, the limitations (RQ4) and whether the system explains its recommendations. A large language model produced a first classification at screening, every decision and value was then checked by all of the authors.

2.3 Limitations of the Review

Only two databases were searched, and requiring *maintenance* in the title may have missed studies that mention it only in the abstract. The iterative refinement of the criteria may introduce some selection bias, although all criteria were applied uniformly and are documented in Table 1. Finally, 47 reports could not be retrieved, and the use of a language model was mitigated by having every output verified by the authors.

3 Background: Maintenance Strategies

Maintenance strategies differ in what triggers an intervention and in the information they require (Table 2, Appendix A). **Corrective** maintenance restores equipment after a failure; it needs no planning but exposes the plant to unplanned downtime and safety risks. **Preventive** maintenance acts at fixed time or usage intervals, which is easy to plan but ignores the state of each machine, so some components are replaced too early and others fail first. **Condition-based** maintenance (CBM) intervenes when a measured indicator, such as vibration or temperature, crosses a threshold. **Predictive** maintenance goes further and estimates the remaining useful life (RUL) with data-driven, physics-based or hybrid models. A RUL estimate, however, does not say which of several pending interventions should come first, nor how to weigh failure likelihood

against safety, production or cost; many retrieved studies stopped at this stage and were excluded under E4. **Prescriptive** maintenance closes this gap by recommending the action, its timing and its resources, combining predictions with costs, constraints and expert knowledge. This is the level at which the present work is placed.

Two management frameworks are often combined with these strategies. Reliability-Centred Maintenance (RCM) defines the policy of each asset through an analysis of its functions, failure modes and consequences; its tools, FMEA and criticality analysis, introduced the ranking of assets by severity and likelihood that underlies many of the models in Section 4. Total Productive Maintenance (TPM) involves operators in equipment care and measures performance through Overall Equipment Effectiveness (OEE), bringing the production cost of downtime into maintenance decisions.

4 Prioritisation Criteria and Methods

This section addresses RQ1 and RQ2: the criteria used to rank maintenance interventions in the 31 included studies and the methods that turn them into a priority order or a schedule.

4.1 Prioritisation Criteria

The criteria were grouped into eight categories (Table 3, Appendix A). Only criteria that actually enter the decision model were counted; several authors mention factors they could not use because the data did not exist [13, 24, 27–29].

The most frequent category is not a property of the equipment but the resources available to maintain it: in 19 studies, priority depends on crews, budget, man-hours or downtime windows, so prioritisation is often treated as the allocation of scarce resources rather than as a ranking in isolation. Failure likelihood and production or service impact follow, with 16 studies each. Condition-based criteria appear in 14 studies, all published from 2022 onwards. Safety is the least represented category: it is explicit in only seven studies, while in others it is a level within a broader "consequences" criterion [5, 19] or is deliberately excluded [31].

4.2 Criticality Analysis and FMEA

Criticality analysis ranks assets or failure modes by combining severity and likelihood, usually in a matrix with a few classes. FMEA adds detectability and summarises each failure mode with the Risk Priority Number, $RPN = S \times O \times D$. The RPN is familiar to maintenance teams, but different score combinations give the same value, ordinal scales are multiplied as if they were ratio scales, the three factors weigh the same, and cost and resources are ignored. The classic form is still used in practice [8, 15]. Most other studies modify it: they weight S, O and D through the Analytic Hierarchy Process (AHP) or the Best-Worst Method (BWM) [11, 22, 24], replace the product by a multi-criteria decision-making (MCDM) ranking [11, 18, 21, 32], represent the uncertainty of expert scores with fuzzy sets [2, 22, 24], or update the risk over time, after each maintenance action [22] or as a preventive task is delayed [26]. Hedayatnia et al. [16] cluster a condition index and

an importance index into nine priority segments, a data-driven version of the criticality matrix.

4.3 Multi-Criteria Decision-Making Methods

MCDM methods serve two tasks. For **weighting**, AHP derives weights from pairwise comparisons and checks their consistency [5, 19, 21, 32]; the check matters, since only four of sixteen expert questionnaires were consistent in the group AHP of Rios et al. [24]. BWM, which needs fewer comparisons, was used mainly in power distribution [3, 11, 18]. Other studies assume equal weights [7, 8, 28], tune them against validation metrics [27] or do not report them [11, 29]. For **ranking**, TOPSIS, which orders alternatives by their closeness to an ideal solution, is the most frequent method [3, 5, 18, 19], followed by PROMETHEE II [19, 21]; grey relational analysis [11] and MABAC [27] appear once, and weighted sums are used for simple indices [7, 16, 28]. VIKOR and ELECTRE were not used. Fuzzy variants [2, 3, 21, 22, 29] capture vague judgements and expert disagreement but add many parameters fixed by the authors.

4.4 Optimisation, Scheduling and Machine Learning

A ranking does not say which interventions fit the available resources or when to perform them. Fifteen studies add an optimisation model. Mixed-integer linear programming (MILP) assigns strategies to failure modes under budget, outage and crew constraints [11, 18], schedules asset fleets [25], allocates technicians [12] or plans over a rolling horizon [6]; smaller integer models select what to repair within a man-hour or budget limit [7, 23, 24]. Larger problems use metaheuristics: adaptive large neighbourhood search [28, 30], particle swarm optimisation [4, 9], a multi-objective genetic algorithm [10] and a two-phase time-window algorithm [13]. Li et al. [17] are the only authors to use deep reinforcement learning (DRL). Priority enters these models either as a coefficient of the objective, which allows trade-offs [6, 7, 24], or as a constraint that guarantees an order at the cost of efficiency [9, 10, 12, 13].

Machine learning (ML) mostly estimates an input to the prioritisation, such as failure probability, RUL or a risk score [6, 7, 10, 23, 27, 29]. Astepe and Alkara [1] instead learn the priority directly from past expert decisions, which avoids explicit weights but reproduces any bias in those decisions. Finally, two studies rank machines by their effect on production, through sensitivity analysis on a line simulation [31] or bottleneck detection on manufacturing execution system (MES) data [14]. Most studies chain several of these methods.

5 Decision Support Systems for Maintenance Prioritisation

This section addresses RQ3, on architecture, data and interaction with the decision-maker, and RQ4, on validation.

5.1 Architecture, Data and Interaction

A DSS combines data, models and an interface through which the decision-maker uses the results. The included studies cover these components to very different degrees. Only nine deliver an implemented tool or pipeline, such as an offshore painting prototype with 3D visualisation [23], a MATLAB tool linking diagnosis, prediction and prescription [29], or an end-to-end pipeline from telemetry ingestion to a MILP schedule [6]; the others at this level are [1, 7, 12, 17, 27, 28]. Thirteen propose a framework or layered architecture without a complete system [2, 3, 8–10, 13–15, 18, 21, 24, 26, 32], and nine contribute only a method [4, 5, 11, 16, 19, 22, 25, 30, 31]. Described architectures follow the same layers (data acquisition, analysis, decision and presentation), and only five studies include a feedback loop or rolling horizon that revises priorities after interventions [6, 8, 14, 15, 24].

The data come from expert judgement, computerised maintenance management system (CMMS) and enterprise resource planning (ERP) records, MES and PLC production data, condition monitoring and engineering models such as grid simulators. Only three studies use public benchmarks [6, 27, 30], case data are often confidential [6, 17, 24, 29], and some models rely on assumed parameters [10, 13, 25]. Integration with the CMMS is recommended but rarely achieved [8, 11], and data quality is a recurring obstacle: manually recorded CMMS data underestimated stoppages compared with MES data [14], and historical priorities used for training contained errors [1].

Interaction with the decision-maker receives little attention. The output is usually a ranked list, a criticality class or a schedule, at most shown as a Gantt chart or a two-dimensional map [12, 13, 16]. Only Rios et al. [23] let the user compare planning strategies, and acceptance was assessed through focus groups in two studies [14, 23]. Explainability, assessed during extraction, is full in eleven studies [2, 3, 6, 7, 14, 15, 21, 24–26, 32], partial in seventeen and absent in three [4, 17, 30]. It usually comes from the transparency of the method (published weights, rules or objective terms) rather than from a mechanism designed for the user; only Danach et al. [6] link each scheduled intervention to the sensor variables behind it, through SHAP values. Experts take part only in eliciting weights [18, 21, 22, 24], and no system supports a group decision while the plan is made.

5.2 Validation and Industrial Application

All 31 studies use real industrial data, as required by E10, mainly from energy (11 studies), manufacturing (11) and oil and gas (6), with single cases in construction equipment, mining and aviation. Table 4 in Appendix A groups them into four types of validation: field implementation (6 studies), comparison with current practice (11), comparison with other methods (8) and simulation or benchmark (6).

Only six studies report results after applying the recommendations in the plant, and the evidence is limited: a year without failures in critical hydroelectric equipment [5, 19], indicators over a six-month pilot [8], higher availability in a cement mill [15], one executed scenario [29] and three inspected circuit breakers [16]. None uses a control group, so the improvement cannot be separated from other changes. The largest group compares the proposal with current practice on the same data; Rios et al. [24], for

example, allocate about twice as much effort to critical assets with the same man-hours, and Wehbi et al. [28] reduce total service time by about 42%. These comparisons are retrospective, so the effect on failures and costs remains an estimate. Comparisons between methods often use weak baselines, such as first-fault-first-repair [13] or a purely reactive policy [6], and statistical tests are rare [14, 17, 27]. Although all studies use real data, few provide strong evidence that prioritisation improved maintenance outcomes.

6 Critical Analysis, Gaps and Trends

6.1 Comparison of the Approaches

The studies fall into five families according to the method at the core of the prioritisation: MCDM, risk and FMEA, hybrid approaches with ML, optimisation and scheduling, and simulation or industrial practice (Table 5, Appendix A). None is better in every respect. MCDM and risk-based methods are transparent and work with expert knowledge alone, but they produce a static ranking that ignores resources unless coupled with optimisation. Optimisation handles resources and time but usually receives the priorities as given. Hybrid approaches use condition data but depend on large histories and add components the user cannot inspect. Studies grounded in industrial practice measure the effect on production but are specific to one line or plant and are hard to generalise.

6.2 Gaps

Six gaps emerge from the analysis, each pointing to a requirement for a decision support system.

G1. Safety is rarely a separate condition. It is explicit in only seven studies and is usually aggregated with other factors [8, 21], so a low safety score can be offset elsewhere. A DSS should treat safety and regulatory requirements as non-compensable constraints.

G2. Ranking and scheduling are separate. Most MCDM and risk-based studies ignore crews, parts and downtime windows, while some scheduling models take priorities as given [12]; few chain both [6, 7, 18, 24]. A DSS should derive priorities and plan interventions in the same process.

G3. Priorities are rarely updated. Only five studies revise priorities through a feedback loop or rolling horizon. A DSS should recompute them when new condition data, failures or work orders appear.

G4. Weights lack a solid basis. They are assumed equal, unreported or taken from few experts [8, 19, 29], and sensitivity analysis is rare [3, 7, 24, 28]. A DSS should make weights explicit, editable and tested for sensitivity.

G5. Explanation and collaboration are limited. Twenty studies explain their recommendations only partially or not at all, only one links each decision to its data [6], and no system supports a joint decision between maintenance, production and safety. A DSS should justify each priority by criterion and let several stakeholders review the plan.

G6. Evidence of real impact is weak. Only six studies applied their results in the plant, always without a control group. A DSS should record decisions and outcomes so that its effect can be measured over time.

6.3 Trends

Nine of the 31 studies date from 2026, including five of the seven hybrid ML studies, against one in 2021 [1]. Four trends stand out: prediction feeding optimisation directly [6, 7, 10, 28]; prescriptive systems that generate and rank alternative actions [29]; explainability through SHAP values or decomposed risk scores [6, 27]; and digital twins and IoT architectures as the data layer [9, 10], together with learning-based scheduling [17]. The most recent proposals are also the most complex, and none has yet been evaluated over a long period of real operation.

7 Proposal

We propose a decision support system that helps the maintenance manager decide the order in which pending interventions should be carried out. The system combines the predicted failure probability of each asset with a multi-criteria decision method and returns a justified ranking. Its design follows the gaps identified in Section 6: safety is a non-compensable constraint (G1), criteria weights are elicited from several stakeholders through a structured method and can be tested for sensitivity (G4), and every position in the ranking is explained in terms of the criteria (G5). The ranking is recomputed when new condition data or failures arrive, which partly addresses G3, and the history of decisions is stored so that their outcomes can later be assessed (G6). Joint ranking and scheduling of interventions (G2) is left for future work.

7.1 System Context

The main user is the maintenance manager or planner, while production and safety managers take part in defining the importance of the criteria. The problem is semi-structured: part of the information is quantitative (condition data, costs, repair times) and part depends on expert judgement. The ranking is produced asynchronously, at the start of each shift, and recomputed whenever an urgent failure is reported; the definition of weights is a synchronous group activity, repeated when priorities in the plant change. Safety risk is a mandatory criterion that cannot be relaxed, so interventions with a risk to people are always placed first. Failure probability, production impact, cost, intervention duration and availability of technicians and spare parts can be relaxed and compensate each other. Each stakeholder group states its own judgements, which the system aggregates while showing where they diverge. Condition data (voltage, rotation, pressure and vibration in the prototype) feed the failure prediction. The interface is a web application that shows the ranking, the explanation of each position and a what-if view in which the user changes the weights and sees the effect on the ranking.

7.2 Requirements

Table 6 (Appendix A) lists eight functional (FR) and five non-functional (NFR) requirements and the gap from Section 6.2 that each one addresses. The functional requirements cover data import, failure prediction, group elicitation of weights with a consistency

warning, the safety constraint, ranking and re-ranking, explanation, sensitivity analysis and the record of decisions; the non-functional ones concern performance, usability, transparency, interoperability and security.

7.3 Conceptual Architecture

The architecture (Fig. 2, Appendix A) has five modules. The **data module** collects condition data, work orders, costs and stock levels. The **predictive module** estimates the failure probability of each asset with a machine learning model. The **weighting module** collects the pairwise comparisons of each stakeholder, checks their consistency and aggregates them into a single set of AHP weights. The **decision module** first separates the interventions that involve a safety risk and then ranks all interventions with TOPSIS, keeping the safety-critical ones at the top. The **interface** presents the ranking, the explanation of each position and the sensitivity analysis, and records the decision taken, which feeds back into the data module.

Keeping the predictive and decision modules separate allows the prediction model to be replaced without changing the prioritisation logic, as in the hybrid studies of Section 4. AHP was chosen because it is widely used in the reviewed studies, checks the consistency of judgements and supports group elicitation; the consistency warning in FR03 follows the low share of consistent expert judgements reported in Section 4.3. TOPSIS was chosen because it is the most frequent ranking method in the sample and its result is easy to explain as the closeness of each intervention to an ideal one. Treating safety as a constraint rather than a weight prevents a risk to people from being offset by gains in other criteria, which is the main weakness found in the compensatory models reviewed.

7.4 Dataset

The predictive module will be developed on the Microsoft Azure Predictive Maintenance dataset[1], a public, simulated dataset. It covers 100 machines of four models, aged 0 to 20 years, and contains hourly averages of voltage, rotation, pressure and vibration for 2015 (876,100 records), 3,919 non-critical errors of five types, 3,286 replacements of four components and 761 failures. The data have no missing values or duplicates.

The exploratory analysis (Appendix B) shows that each component has its own sensor signature about 48 hours before a failure, that failures are more frequent in older machines and in machines with more errors, and that only 2% of machine-hours precede a failure within 24 hours, which implies strong class imbalance.

In the prototype, each pending intervention is the replacement of one component in one machine. The predictive module estimates the probability that the component fails within the next 24 hours from rolling sensor statistics, error counts, time since the last replacement, and machine age and model. Cost, duration, production impact and safety risk are not in the data and will be assigned per component and machine model as stated assumptions. Because the data are simulated, the prototype can show that the system works as designed, but not that it improves maintenance in a real plant; that evaluation remains future work (G6).

[1] `kaggle.com/datasets/arnabbiswas1/microsoft-azure-predictive-maintenance`

8 Conclusions

This review analysed 31 studies, published from 2021 onwards, that prioritise industrial maintenance interventions and evaluate their approach in a real setting. The building blocks are mature: criteria centred on resources, failure likelihood and production impact, weighting and ranking methods such as AHP, BWM and TOPSIS, optimisation models for crews, budgets and time windows, and increasingly reliable failure predictions. What remains open is how these blocks are put together. Safety is seldom treated as a separate condition, ranking and scheduling are usually solved apart, priorities are rarely updated, weights have a weak basis, recommendations are only partly explained, and few studies show real improvements in maintenance results.

The proposed DSS addresses part of these gaps by combining failure prediction with AHP and TOPSIS, placing safety outside the trade-off between criteria and explaining every position in the ranking. The next steps are to implement the prototype on the Microsoft Azure dataset characterised in Section 7.4 and Appendix B, elicit the weights with maintenance, production and safety staff, and evaluate it in scenarios, comparing its ranking with that of an experienced planner, testing its sensitivity to the weights and measuring whether users accept the explanations. Integration with scheduling and validation with real plant data are left for later stages.

In our view, as digital twins and condition monitoring make more data available, the value of DSS in industrial maintenance will lie less in replacing the planner than in making decisions explicit: showing why an intervention comes first, helping departments agree on that order and keeping a record from which the plant can learn.

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A Supplementary Tables and Figures

This appendix gathers supporting tables and figures referred to in Sections 2 to 7.

Table 1. Exclusion criteria.

Code	Criterion
E1	Duplicate record.
E2	Not a peer-reviewed primary study (review, survey, editorial or preprint).
E3	Outside the industrial context (e.g., healthcare, buildings, civil infrastructure, IT maintenance).
E4	Only failure prediction, detection or diagnosis, with no decision component.
E5	No evaluation of any kind.
E6	Insufficient methodological detail.
E7	Full text not accessible.
E8	Maintenance management without decisions on interventions or assets (e.g., skills, barriers, KPIs).
E9	The decision is not the prioritisation, selection or scheduling of interventions or assets (e.g., strategy selection, interval optimisation, joint production and maintenance scheduling).
E10	Not applied to real data or an industrial case with a comparison against current practice, another method or a before/after measurement.

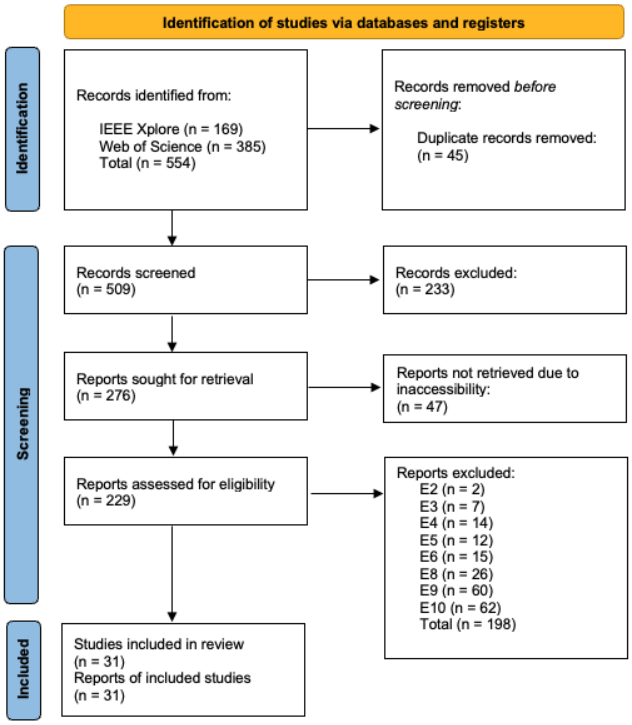


Fig. 1. PRISMA 2020 flow diagram of the study selection.

Table 2. Comparison of maintenance strategies.

Strategy	Trigger	Information required	Main limitation
Corrective	Occurrence of a failure	Failure reports	Unplanned downtime and secondary damage
Preventive	Fixed time or usage interval	Manufacturer data, failure history	Interventions on healthy equipment
Condition-based	Indicator exceeds a threshold	Condition monitoring data	No anticipation of the failure time
Predictive	Estimated RUL	Sensor data, prognostic models	Does not define which action to take or in which order
Prescriptive	Recommended action and timing	Predictions, costs, resources and constraints	Requires integration of data, models and organisational knowledge

Table 3. Prioritisation criteria used in the included studies (n = number of studies).

Category	Typical indicators	Studies	*n*
Resources and operational constraints	Crews, skills, man-hours, budget, spare parts, downtime windows, travel	[4–7,10–13,17–19,21,23–25,28–30,32]	19
Likelihood of failure	Occurrence, failure rate, MTBF, number of past failures	[1,2,5–9,11,14,15,18,19,21,22,24,32]	16
Production or service impact	Downtime, bottlenecks, OEE, line efficiency, energy not supplied, SAIFI, SAIDI	[3,6,9–11,13–16,18,21,24,26,29,31,32]	16
Cost	Intervention cost, downtime and interruption cost, financial risk	[2,3,6,7,9–11,13,18,25,26,28–30,32]	15
Condition, degradation and RUL	Sensor-based health indices, wear, corrosion, predicted RUL	[4,6,7,10,13,16,23–29,32]	14
Detectability	Detection of the failure before it causes damage	[5,8,11,15,18,19,21,22,24]	9
Asset importance	Criticality class, importance measures, regulatory requirements	[1,9,10,12,13,16,18,23,29]	9
Safety and environment	Risk to people, explosion, environmental damage	[2,8,11,15,21,24,26]	7

Table 4. Types of validation in the included studies (n = number of studies).

Type	Description	Studies	n
Field implementation	Recommendations applied in the plant; indicators compared before and after	[5,8,15,16,19,29]	6
Comparison with current practice	Proposed plan compared with the company's plan or with expert decisions on the same data	[1,2,4,11,12,17,18,23,24,26,28]	11
Comparison with other methods	Same case solved with alternative methods or policies	[3,6,7,13,21,22,27,32]	8
Simulation or benchmark	Simulation model of the real system or public benchmark instances	[9,10,14,25,30,31]	6

Table 5. Comparison of the families of approaches found in the included studies.

Family	Strengths	Limitations	Studies
MCDM	Transparent; uses expert knowledge; needs no sensor data	Static ranking; subjective weights; compensatory aggregation; resources only when combined with optimisation	[3, 5, 11, 18, 19, 21, 24]
Risk and FMEA	Familiar to maintenance teams; consistent with RCM; links severity and likelihood	Ordinal scores; ranks risks rather than interventions; resources rarely considered	[2, 16, 22, 26, 32]
Hybrid with ML	Uses condition data; can be updated as new data arrive	Needs large labelled histories; labels are proxies or past decisions; predictive part hard to explain	[1, 6, 7, 23, 27–29]
Optimisation and scheduling	Handles crews, budgets and time windows; scales to large instances	Priorities given a priori or hidden in costs; many assumed parameters; metaheuristics and DRL hard to explain	[4,9,10,12,13,17,25,30]
Simulation and industrial practice	Grounded in production impact; implemented in real plants	Specific to one line or plant; hard to generalise; evaluation without control	[8, 14, 15, 31]

Table 6. Functional and non-functional requirements of the proposed system.

ID	Requirement	Gap
FR01	Import pending interventions, equipment data and work orders (e.g., from CSV files or a CMMS).	–
FR02	Estimate the failure probability of each asset from its condition data.	–
FR03	Let each stakeholder define the importance of the criteria through pairwise comparisons, warn when judgements are inconsistent ($CR \geq 0.1$) and aggregate the judgements of all stakeholders.	G4, G5
FR04	Place every intervention with a safety risk at the top of the list, regardless of the other criteria.	G1
FR05	Rank the remaining pending interventions and recompute the ranking when new data or failures arrive.	G3
FR06	Explain each position by showing the contribution of each criterion.	G5
FR07	Show how the ranking changes when the weights are changed (sensitivity analysis).	G4
FR08	Store each ranking, the decision taken and, later, its outcome.	G6
NFR01	Performance: produce the ranking in a few seconds for a few hundred interventions.	–
NFR02	Usability: be understandable by users without knowledge of MCDM methods.	–
NFR03	Transparency: no recommendation is shown without its justification.	G5
NFR04	Interoperability: import data in common formats and allow connection to a CMMS.	–
NFR05	Security: require authentication, with separate profiles for managers and other users.	–

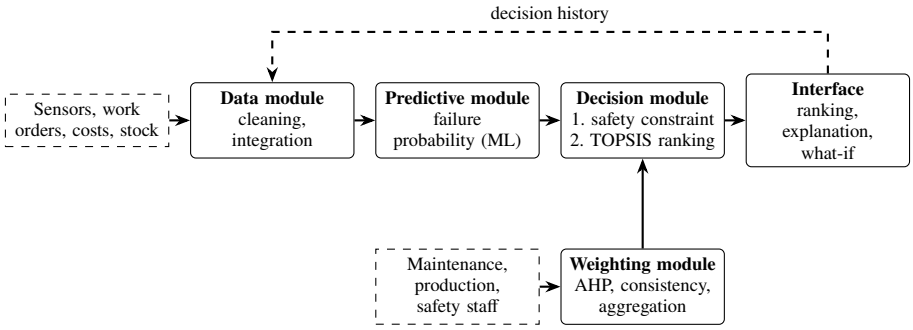


Fig. 2. Conceptual architecture of the proposed system.

B Exploratory Analysis of the Dataset

This appendix characterises the Microsoft Azure Predictive Maintenance dataset described in Section 7.4. The analysis was carried out in Python with pandas and matplotlib.

Structure and quality. The dataset is split into five files (Table 7). All 100 machines have a complete hourly telemetry series for 2015 (8,761 records each), and no file contains missing values or duplicated rows. The maintenance file also includes 400 replacements from 2014, which provide the history needed to compute the time since the last replacement at the start of 2015. Of the 761 failures, 743 match a replacement in the maintenance file; the remaining 18 all occur on 2 January 2015 at 03:00 and are an artefact of the start of the series rather than missing records. Failures account for 25.7% of the 2,886 replacements made in 2015, so about three quarters of the interventions are preventive.

Table 7. Files of the Microsoft Azure Predictive Maintenance dataset.

File	Records	Content
Telemetry	876,100	Hourly mean of voltage, rotation, pressure and vibration per machine, 1 January 2015 to 1 January 2016
Errors	3,919	Non-critical errors of five types (error1–error5), with timestamp and machine
Maintenance	3,286	Component replacements (comp1–comp4), scheduled or after failure, June 2014 to December 2015
Failures	761	Replacements caused by a failure, by component, in 98 of the 100 machines
Machines	100	Model (model1–model4) and age in years (0–20)

Sensor distributions. The four sensors follow approximately symmetric, bell-shaped distributions (Fig. 3), with means of 170.8 (voltage), 446.6 (rotation), 100.9 (pressure) and 40.4 (vibration). The correlation between any two sensors is close to zero, so each carries independent information. Extreme readings are concentrated before failures: between 18% and 26% of the readings more than three standard deviations from the mean fall in the 48 hours that precede a failure, although these periods cover only 4% of the data.

Failures by component and machine model. Component 2 fails most often (259 failures) and component 3 least often (131). Model 3 and model 4 account for two thirds of the fleet, but model 1 and model 2 machines fail about twice as often per machine (Fig. 4). The median interval between two failures of the same component in the same machine is 75 days. Monthly failures are stable, between 50 and 69, except for January, which is inflated by failures recorded on the first day of the series.

Behaviour before failure. Fig. 5 shows, for each component, the deviation of one sensor from its overall mean in the 96 hours before a failure. Each component is associated with a different sensor, and the deviation appears abruptly about 48 hours before the failure: voltage rises by about 12% before failures of component 1, rotation falls by about 17%

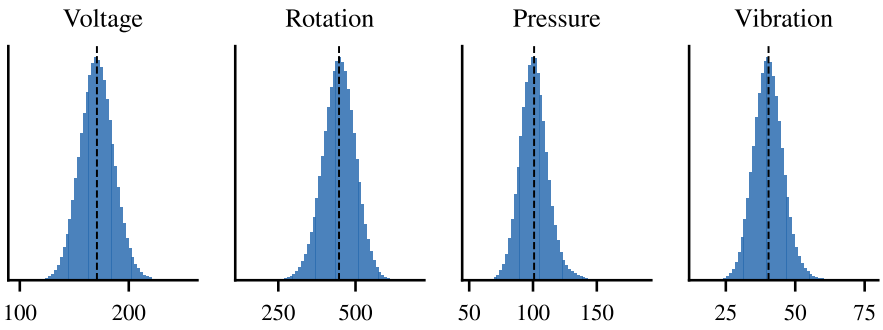


Fig. 3. Distribution of the hourly sensor readings (dashed line: mean).

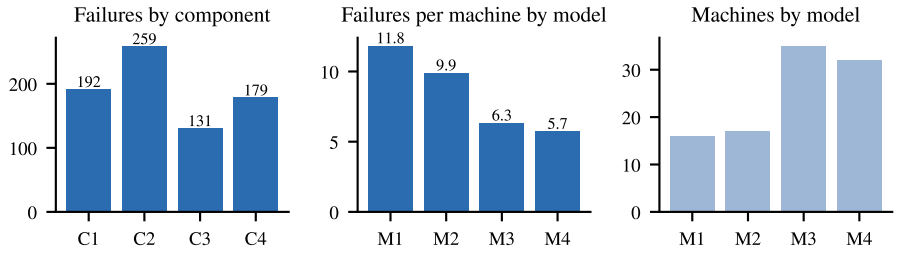


Fig. 4. Failures by component, failures per machine by model, and number of machines by model.

for component 2, pressure rises by about 24% for component 3 and vibration rises by about 25% for component 4. This pattern supports a classifier based on rolling sensor statistics over the last 24 to 48 hours. At the same time, only 1.96% of machine-hours precede a failure within 24 hours, so the training will have to handle a strong class imbalance.

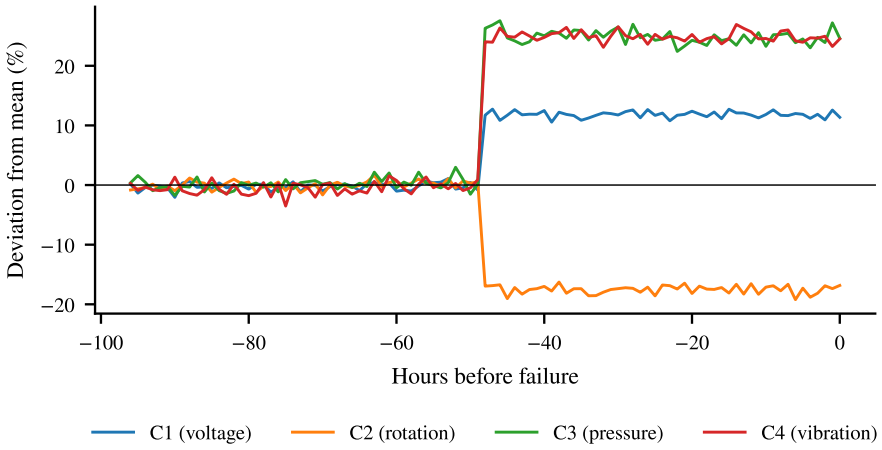


Fig. 5. Mean deviation of the associated sensor from its overall mean before failures of each component.

Machine age and errors. The number of failures per machine grows with machine age ($r = 0.47$) and with the number of errors recorded during the year ($r = 0.54$), as shown in Fig. 6. Errors are therefore useful as an early indicator, even though they do not stop the machine, and age and model can serve as measures of asset importance in the ranking.

Implications for the prototype. The dataset provides what the predictive module needs: a clear signal before failures, an identifiable set of assets and components, and a maintenance history. It does not provide costs, intervention durations, production impact or safety information, which will be added as assumptions. Since the data are simulated, the prototype will demonstrate the behaviour of the system rather than its effect on a real plant.

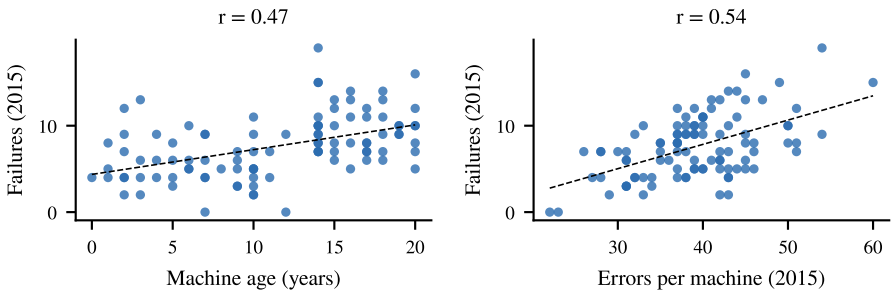


Fig. 6. Failures per machine in 2015 against machine age and against the number of errors.